Study Environment monitoring

Presented by FinalBossMaYuen

OVERVIEW

Environmental factors such as lighting, noise, temperature, humidity, and air quality can significantly affect students' concentration and productivity, but they are often not monitored.

With the advancement of Internet of Things (IoT) technology and web APIs, it is now possible to collect real-time environmental data and external information like weather and air quality.

However, there is a lack of systems that combine these data to evaluate study environments. Therefore, this project develops a Study Environment Monitor to assess conditions and provide feedback to improve study effectiveness.

ARCHITECTURE

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MQTT X

React

Tailwind CSS

phpMyAdmin

node JS

TECH STACK

Frontend:

- Next.js 16 + React 19
- TypeScript
- Tailwind CSS 4
- Chart.js + react-chartjs-2

Backend:

- Node.js + Express 5
- MySQL + mysql2
- TensorFlow.js
- Swagger UI / OpenAPI 3.0

DATA SOURCES

Primary :

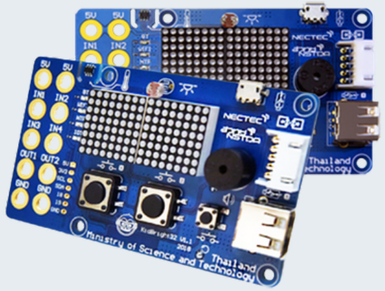
- PIR
- Big Sound Detector
- Photoresistor
- Temperature and humidity

Secondary :

- OpenWeather API
- OpenAQ API
- IQAir AirVisual API

COLLECTION METHOD

- Kidbright Board wired with sensors



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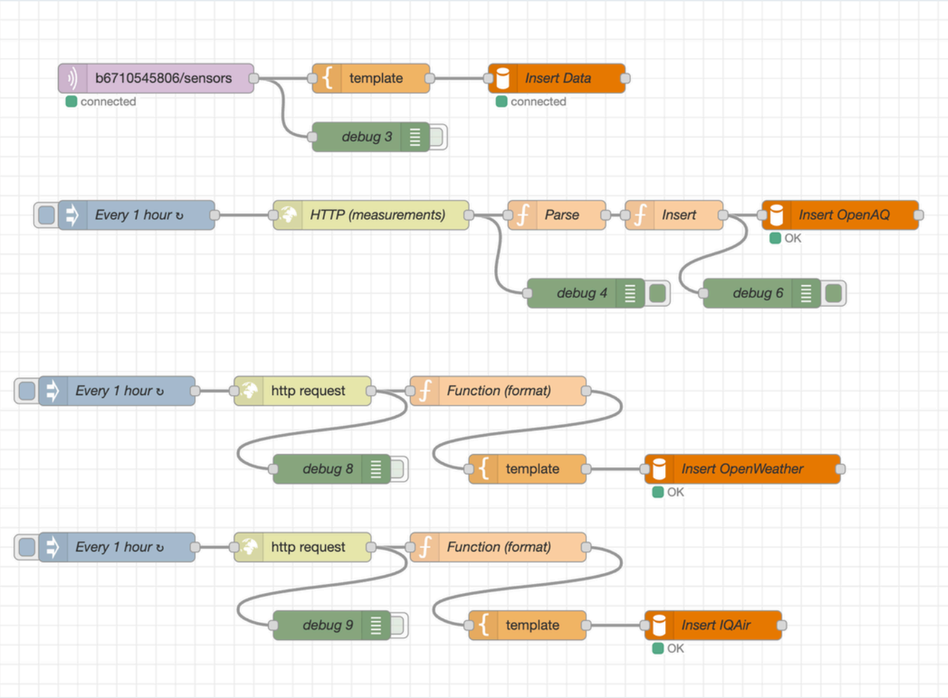
COLLECTION METHOD

- Node-Red



DATABASE SCHEMA

sensor data

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	bigint			No	*None*		AUTO_INCREMENT	Change Drop More
2	**pir_motion**	tinyint(1)			Yes	*NULL*			Change Drop More
3	**light_level**	float			Yes	*NULL*			Change Drop More
4	**temperature**	float			Yes	*NULL*			Change Drop More
5	**humidity**	float			Yes	*NULL*			Change Drop More
6	**noise_level**	float			Yes	*NULL*			Change Drop More
7	**timestamp**	datetime			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More

study index

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	bigint			No	*None*		AUTO_INCREMENT	Change Drop More
2	**pir_score**	float			Yes	*NULL*			Change Drop More
3	**light_score**	float			Yes	*NULL*			Change Drop More
4	**temp_score**	float			Yes	*NULL*			Change Drop More
5	**humidity_score**	float			Yes	*NULL*			Change Drop More
6	**noise_score**	float			Yes	*NULL*			Change Drop More
7	**aqi_score**	int			No	*None*			Change Drop More
8	**total_score**	float			Yes	*NULL*			Change Drop More
9	**status**	varchar(20)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
10	**timestamp**	datetime			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More

DATABASE SCHEMA

openweather

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	bigint			No	*None*		AUTO_INCREMENT	Change Drop More
2	**temperature**	float			Yes	*NULL*			Change Drop More
3	**humidity**	float			Yes	*NULL*			Change Drop More
4	**pressure**	float			Yes	*NULL*			Change Drop More
5	**wind_speed**	float			Yes	*NULL*			Change Drop More
6	**weather_main**	varchar(50)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
7	**weather_desc**	varchar(100)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
8	**timestamp**	datetime			Yes	*NULL*			Change Drop More

openaq

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	int			No	*None*		AUTO_INCREMENT	Change Drop More
2	**sensor_id**	int			No	*None*			Change Drop More
3	**value**	float			No	*None*			Change Drop More
4	**units**	varchar(20)	utf8mb3_general_ci		Yes	µg/m³			Change Drop More
5	**aqi_label**	varchar(50)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
6	**timestamp_local**	datetime			Yes	*NULL*			Change Drop More
7	**timestamp_utc**	datetime			Yes	*NULL*			Change Drop More
8	**has_flags**	tinyint(1)			Yes	0			Change Drop More
9	**created_at**	timestamp			Yes	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More

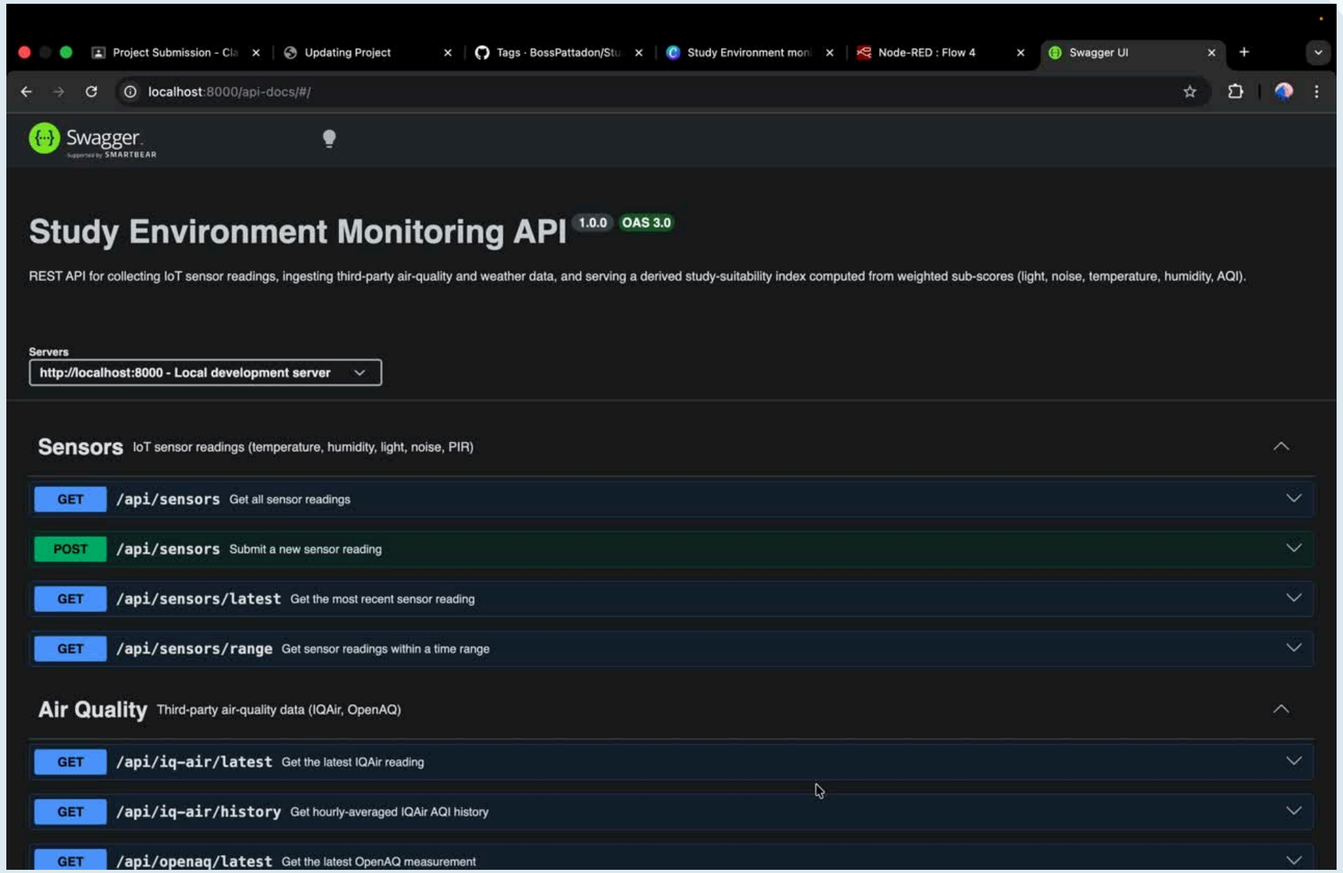
DATABASE SCHEMA

iq_air

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	bigint			No	*None*		AUTO_INCREMENT	Change Drop More
2	**aqi_us**	int			Yes	*NULL*			Change Drop More
3	**main_pollutant**	varchar(20)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
4	**pm25**	float			Yes	*NULL*			Change Drop More
5	**pm10**	float			Yes	*NULL*			Change Drop More
6	**o3**	float			Yes	*NULL*			Change Drop More
7	**no2**	float			Yes	*NULL*			Change Drop More
8	**so2**	float			Yes	*NULL*			Change Drop More
9	**co**	float			Yes	*NULL*			Change Drop More
10	**temperature**	float			Yes	*NULL*			Change Drop More
11	**humidity**	float			Yes	*NULL*			Change Drop More
12	**pressure**	float			Yes	*NULL*			Change Drop More
13	**wind_speed**	float			Yes	*NULL*			Change Drop More
14	**wind_direction**	int			Yes	*NULL*			Change Drop More
15	**city**	varchar(100)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
16	**country**	varchar(100)	utf8mb3_general_ci		Yes	*NULL*			Change Drop More
17	**timestamp**	datetime			Yes	*NULL*			Change Drop More
18	**created_at**	datetime			Yes	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More

DATA SHARING API

Swagger™
Supported by SMARTBEAR



STUDY INDEX SCORE CALCULATION

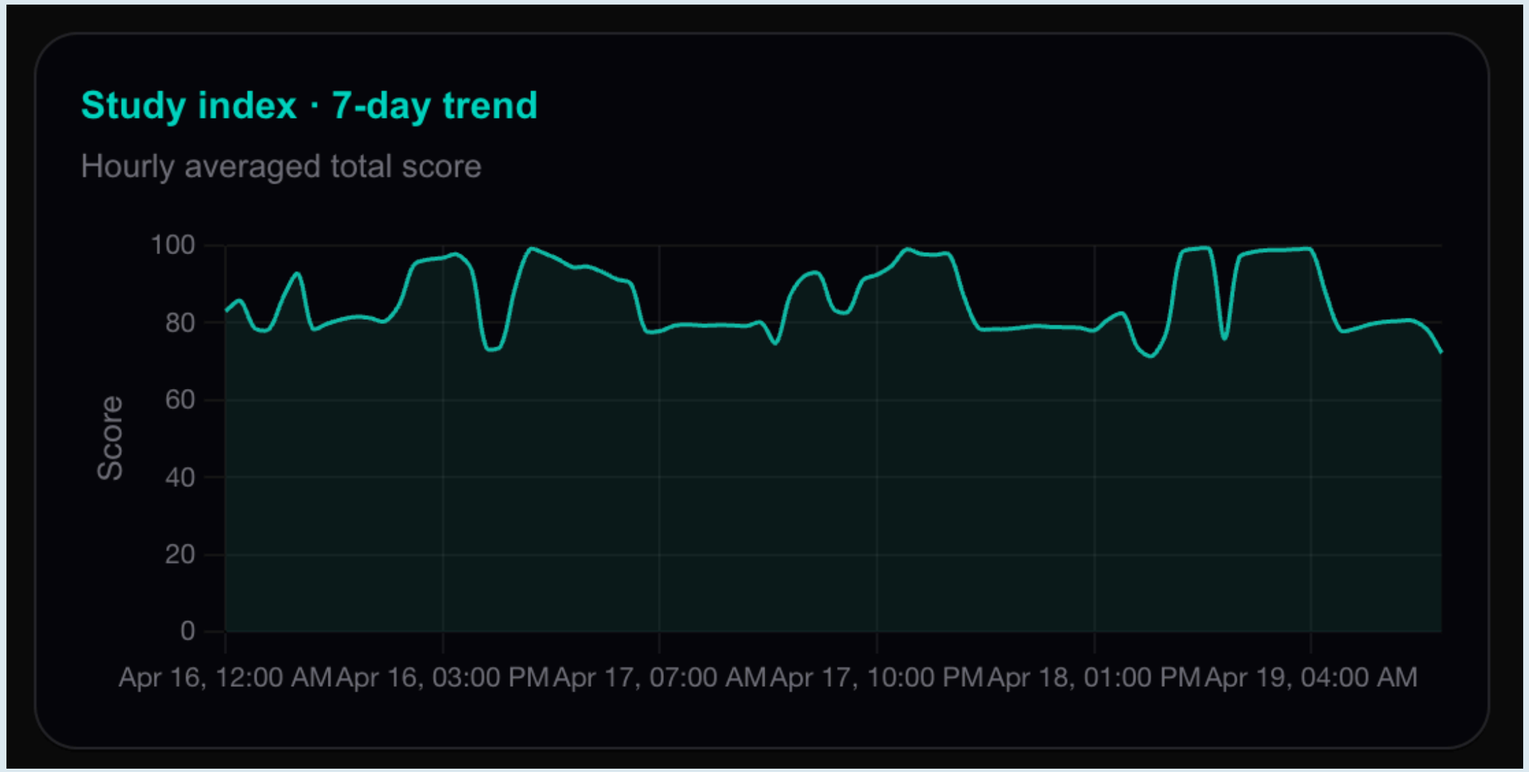
Final Score:

total = Σ(sub_score × weight) / Σ(weights)

Weights are user-configurable (default:
Noise 30%, Light 25%,
Temp 25%, Humidity 10%, Air 10%)

Factor	Ideal Range	Penalty Logic
Light	300–500 lux	Linear ramp to 100 below 300; gentle -25 pts from 500–1000; steeper penalty above 1000
Noise	ADC ⩽ 620	100 below 620; linear drop to 0 at 900
Temperature	20–23°C	-5 pts/°C below ideal; -7 pts/°C above
Humidity	40–60% RH	-4 pts/% below; -3 pts/% above
AQI	⩽ 50	Piecewise: 100→75 (50–100 AQI), 75→50 (100–150), 50→25 (150–200), 25→0 (200–300)

DATA VISUALIZATION



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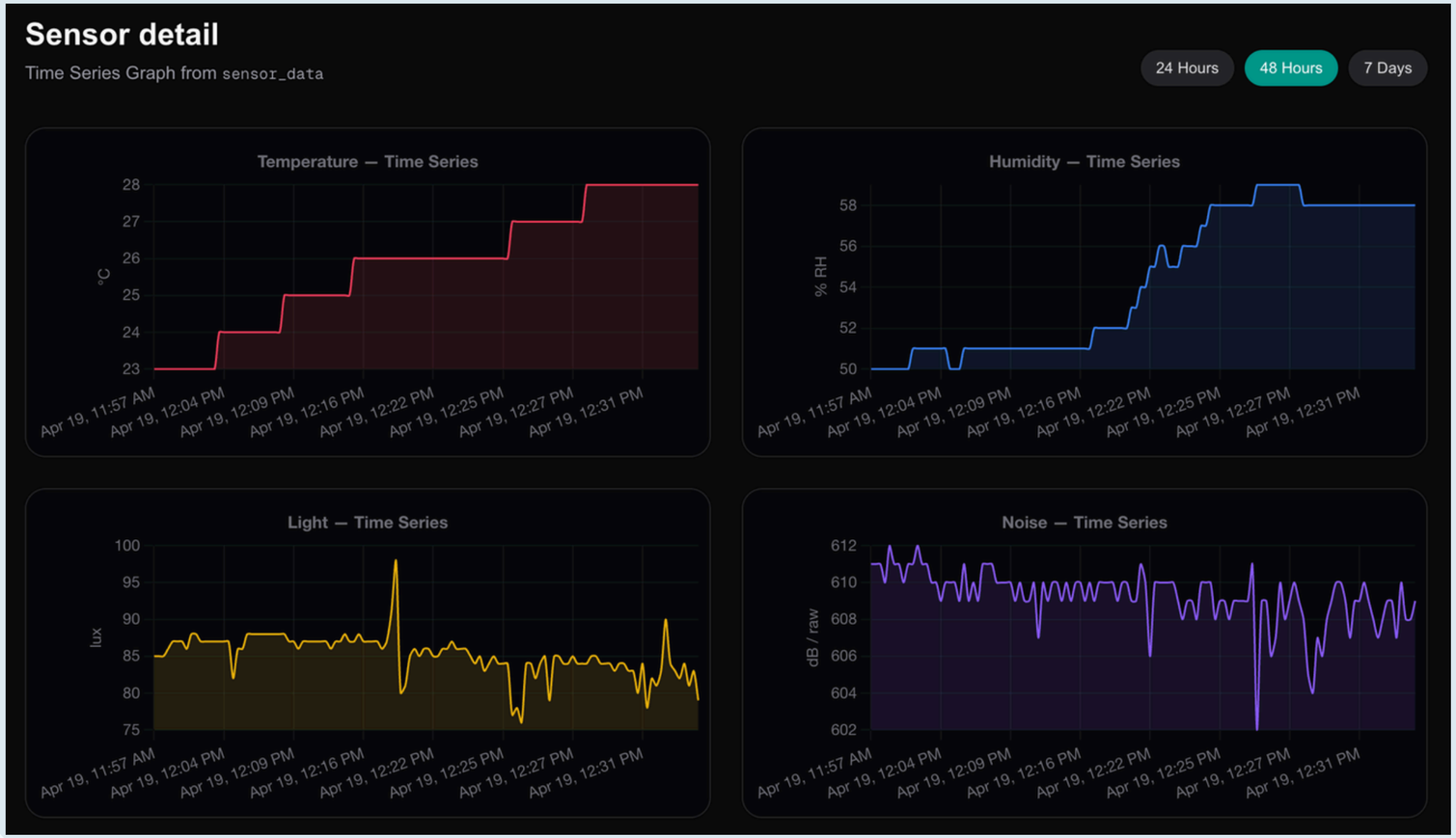

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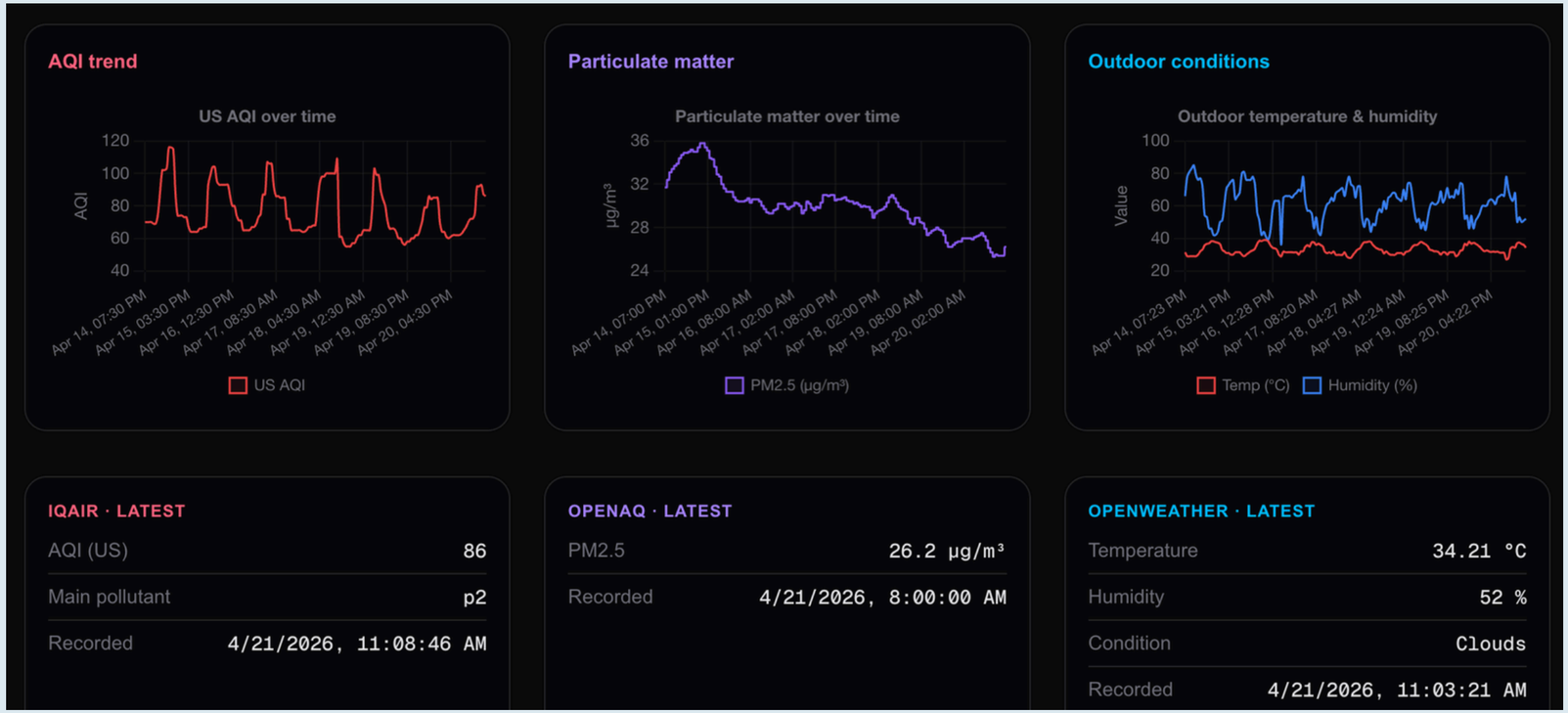

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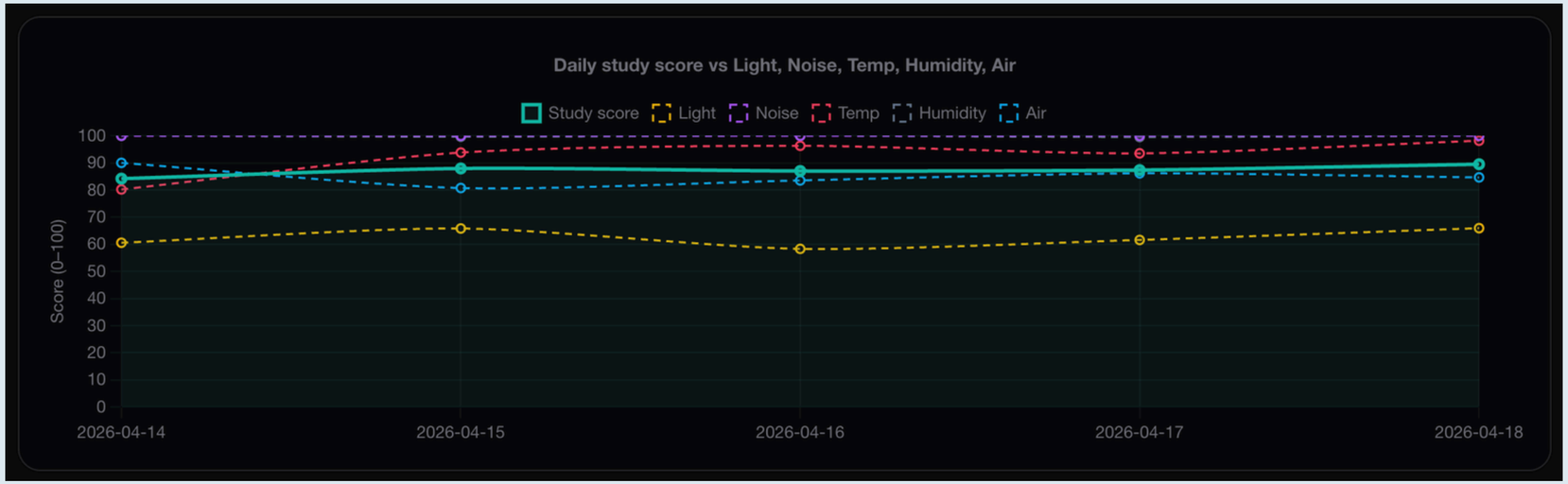

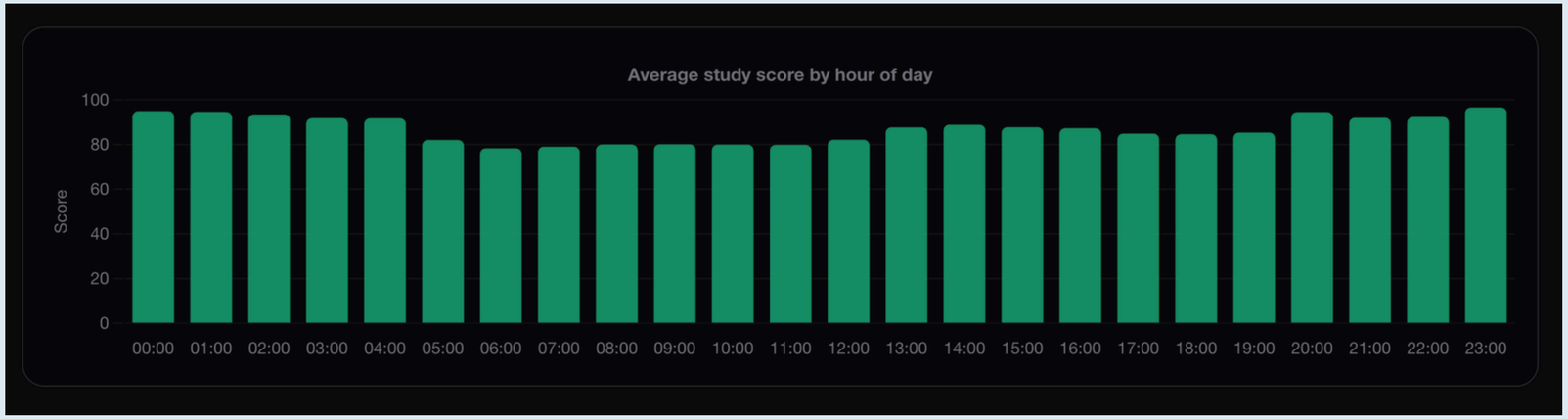

DEMONSTRATION OF KEY FEATURES

1. Live Dashboard
real-time study suitability index (0–100) ring with per-metric sub-scores (light, noise, temperature, humidity, AQI)

2. Air Quality Page
current US AQI, PM2.5, outdoor temp/humidity with time-series charts from IQAir, OpenAQ, and OpenWeatherMap

3. History Page
interactive historical charts and raw data table with configurable date range

4. Forecast Page
ML-based predictions of study index and PM2.5 for the next N hours, with 4 swappable models (MLP, Linear Regression, Decision Tree, Random Forest)

DEMONSTRATION OF KEY FEATURES

5. Insights Page
rule-based recommendations based on current sensor readings

6. Settings Page
configurable scoring weights (light, noise, temp, humidity, AQI)
persisted on the backend

7. Swagger UI
interactive API documentation at /api-docs

MEMBER

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